

LOW ACRYLONITRILE NBR: PATENT LANDSCAPE FOR COMPOSITION CONTROL, MOLECULAR-WEIGHT REDUCTION, AND DOWNSTREAM FORMULATION USE

1. ABSTRACT

This report synthesizes four primary patents relevant to low-acrylonitrile nitrile butadiene rubber (NBR) and closely related low-viscosity nitrile rubber technologies. The target is interpreted as NBR grades with reduced acrylonitrile content and/or reduced molecular weight that improve processability while retaining the oil-, heat-, and chemical-resistance profile of nitrile elastomers. The patent landscape spans two distinct technology families: (1) direct or indirect modification of nitrile rubber by metathesis and hydrogenation to lower molecular weight and narrow dispersity, and (2) downstream compounding of low-viscosity or hydrogenated nitrile rubber into swellable or degradable elastomeric articles. EP2473279B1 and EP2057205B1 are the most synthesis-relevant. EP2473279B1 discloses metathesis of nitrile rubber in the presence of a transition-metal catalyst, followed by hydrogenation in monochlorobenzene using hydrogen pressure and a rhodium catalyst, yielding lower-Mw HNBR suitable for improved processability. EP2057205B1 discloses segment-interchange metathesis of unsaturated polymers, including HNBR, using Grubbs II catalyst or tungsten hexachloride/tri-n-butylmethyltin systems to form meta-block copolymers; while not a direct NBR synthesis route, it demonstrates how HNBR can be chemically reorganized into higher-value architectures. US10385647B2 and US10308807B2 are formulation/application patents rather than synthesis patents: the former uses nitrile elastomer/HNBR in swellable packers with powder superabsorbent polymer, while the latter uses NBR in dissolvable elastomeric composites with water-soluble binders. Across the set, the dominant process trends are emulsion-derived nitrile rubber feedstocks, metathesis-based molecular-weight reduction, selective hydrogenation, solvent-based processing, and compounding strategies that exploit low-viscosity nitrile rubber for injection molding, swellable packers, or degradable seals. The evidence supports low-Mw, low-viscosity nitrile rubber as the most practical route to improved low-acrylonitrile NBR performance, with composition control and post-polymerization modification being the key enabling levers.

2. METHODOLOGY

PRIMARY PATENT EVIDENCE

Patent 1

- Patent Number: EP2473279B1
- Patent Title: Metathesis of a nitrile rubber in the presence of a transition metal catalyst
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2014
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses metathesis degradation of nitrile rubber in solution using a transition-metal catalyst, followed by hydrogenation to produce hydrogenated nitrile rubber with reduced molecular weight and improved processability. The examples show dissolution of nitrile rubber in monochlorobenzene, addition of 1-hexene and a ruthenium metathesis catalyst, then hydrogenation with tris(triphenylphosphine)rhodium chloride under hydrogen pressure. - Nitrile rubber charge: 700 g — Examples 5-6 - Solvent charge: 4667 g monochlorobenzene — Examples 5-6 - Reactor volume: 10 L high pressure reactor — Examples 5-6 - Co-olefin charge: 4 phr 1-hexene — Examples 5-6 - Metathesis temperature: 22°C — Examples 5-6 - Metathesis time: 2 h — Examples 5-6 - Hydrogenation catalyst: 0.06 phr tris-(triphenylphosphine) rhodium chloride — Examples 5-6 - Hydrogen pressure: 85 bar — Examples 5-6 - Hydrogenation temperature: 138°C — Examples 5-6 - Hydrogenation time: 4 h — Examples 5-6 - Agitation speed: 600 rpm — Examples 5-6 - Hydrogenation level: <0.9% — Examples 5-6 - Metathesis catalyst loading: 0.1 phr — Table 3, Example 7 - Solvent content: 1,300 ppm — Table 3, Example 7 - Molecular weight Mn: 19,000 g/mol — Table 3, Example 7 - Molecular weight Mw: 34,000 g/mol — Table 3, Example 7 - PDI: 1.78 — Table 3, Example 7 - Control Mn: 69,000 g/mol — Table 3, Control #1 - Control Mw: 217,500 g/mol — Table 3, Control #1 - Control PDI: 3.15 — Table 3, Control #1

Relevant Experimental Evidence - Example 1: low catalyst loading produced reduced molecular weights relative to control. - Example 2: increasing catalyst loading further reduced molecular weight and PDI. - Example 3: higher catalyst loading continued the trend toward lower Mn and Mw. - Example 4: the highest listed catalyst loading gave the lowest molecular weights among Examples 1-4. - Example 5-7: metathesis followed by hydrogenation produced low-Mw HNBR; Example 7 gave Mn 19,000 g/mol and Mw 34,000 g/mol.

Technical Relevance This is the most directly relevant patent for low-acrylonitrile NBR because it shows how nitrile rubber can be chemically modified to lower molecular weight and viscosity while preserving the nitrile rubber backbone. The disclosed metathesis/hydrogenation route is a practical pathway to low-viscosity NBR/HNBR feedstocks for downstream compounding.

Patent 2

- Patent Number: EP2057205B1
- Patent Title: Production of meta-block copolymers by polymer segment interchange
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2017
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses metathesis-driven segment interchange between two or more ethylenically unsaturated polymers to form meta-block copolymers. In the examples, HNBR is combined with other unsaturated polymers in toluene and treated with Grubbs II catalyst or tungsten hexachloride/tri-n-butylmethyltin catalyst systems, then isolated by methanol precipitation and solvent extraction. - Polymer charge: 0.25 g each HPBD and HNBR — Example 1 - Solvent: toluene — Example 1 - Temperature: 95°C — Example 1 - Catalyst charge: 8 mg Grubbs II metathesis catalyst — Example 1 - Reaction time: 1 h — Example 1 - Precipitation solvent: methanol — Example 1 - Polymer charge: 0.25 g each unsaturated polyurethane and poly(ethylene-co-butadiene) — Example 2 - Solvent: tetrachloroethane — Example 2 - Temperature: 105°C — Example 2 - Catalyst charge: 9 mg Grubbs II metathesis catalyst — Example 2 - Reaction time: 1 h — Example 2 - Extraction solvent: 30 ml tetrahydrofuran — Example 2 - Polymer charge: 1.05 g fumaryl-modified polycarbonate and 0.45 g poly(ethylene-co-butadiene) — Example 3 - Temperature: 105°C — Example 3 - Catalyst charge: 9 mg Grubbs II metathesis catalyst — Example 3 - Reaction time: 1 h — Example 3 - Polymer charge: 0.25 g fumaryl-modified polycarbonate and 0.25 g poly(ethylene-co-butadiene) — Example 4 - Temperature: 105°C — Example 4 - Catalyst charge: 9 mg Grubbs II metathesis catalyst — Example 4 - Reaction time: 1 h — Example 4 - Polymer charge: 0.25 g each unsaturated poly(ethylene oxide) and poly(ethylene-co-butadiene) — Example 5 - Temperature: 105°C — Example 5 - Catalyst charge: 9 mg Grubbs II catalyst — Example 5 - Reaction time: 1 h — Example 5 - Polymer charge: 0.25 g fumaryl-modified polycarbonate and 0.25 g poly(ethylene-co-octene-co-butadiene) — Example 6 - Temperature: 105°C — Example 6 - Catalyst charge: 9 mg Grubbs II metathesis catalyst — Example 6 - Reaction time: 1 h — Example 6 - Polymer charge: 0.10 g poly(ethylene-co-butadiene) and 0.40 g poly(ethylene-co-octene-co-butadiene) — Example 7 - Temperature: 105°C — Example 7 - Catalyst loading: 8,000 ppm tungsten hexachloride — Example 7 - Co-catalyst loading: 13,000 ppm tri-n-butylmethyltin — Example 7 - Reaction time: 1 h — Example 7 - Polymer charge: 0.25 g each unsaturated polyester and poly(ethylene-co-butadiene) — Example 8 - Temperature: 105°C — Example 8 - Catalyst charge: 9 mg Grubbs II catalyst — Example 8 - Reaction time: 1 h — Example 8 - Polymer charge: 0.25 g SBS and 0.25 g poly(ethylene-co-butadiene) — Example 9 - Temperature: 105°C — Example 9 - Catalyst loading: 8,000 ppm tungsten hexachloride — Example 9 - Co-catalyst loading: 13,000 ppm tri-n-butylmethyltin — Example 9 - Reaction time: 1 h — Example 9 - Polymer charge: 0.25 g unsaturated fumaryl-modified polycarbonate, 0.25 g SBS, and 0.25 g poly(ethylene-co-butadiene) — Example 10 - Temperature: 105°C — Example 10 - Catalyst charge: 9 mg Grubbs II catalyst — Example 10 - Reaction time: 1 h — Example 10 -

Extraction solvent: 50 ml tetrahydrofuran — Example 10

Relevant Experimental Evidence - Example 1: HNBR-containing feed produced a meta-block copolymer, unlike the catalyst-free control. - Example 2: segment interchange was confirmed by FTIR after extraction of unreacted polyurethane. - Example 3: TEM and SAXS showed ordered microphase morphology in the metathesized product. - Example 4: catalyst-enabled metathesis produced nano-scale microphase separation, while the control remained a blend. - Example 7: tungsten hexachloride/organotin catalysis enabled segment interchange in a polyolefin blend system.

Technical Relevance Although not a direct NBR synthesis patent, it is relevant because it demonstrates that HNBR can participate in metathesis-based segment interchange to create new low-viscosity or architecturally modified elastomeric materials. This supports the broader utility of nitrile-rubber-derived feedstocks in advanced polymer modification routes.

Patent 3

- Patent Number: US10385647B2
- Patent Title: Swellable rubber compositions
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2019
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent is primarily a compounding and application disclosure for swellable rubber compositions rather than a synthesis route for NBR. It teaches nitrile elastomer-based swellable packers containing butadiene rubber, sulfur curing agents, silica, and powder superabsorbent polymer, with HNBR or nitrile rubber used as the elastomeric phase in downhole sealing articles. - Powder super absorbent polymer loading: 110 phr — Example 2 - Powder super absorbent polymer loading: 130 phr — Example 2 - Powder super absorbent polymer loading: 150 phr — Example 2 - Powder super absorbent polymer loading: 170 phr — Example 2 - Test temperature: 200°F — Example 2 - Test temperature: 275°F — Example 2 - Brine concentration: 5% by volume — Example 2 - Brine concentration: 16% by volume — Example 2 - Predicted SAP loading: about 205 phr — Example 3 - Predicted SAP loading: about 190 phr — Example 3 - Predicted SAP loading: about 160 phr — Example 3 - Maximum load for medium granular SAP: about 170 phr — Example 3 - Maximum load for large granular SAP: about 150 phr — Example 3 - Composition: 10% NBR rubber (Nipol DN3350) — Example 4 - Composition: 10% HNBR (Zetpol 2030L) — Example 4 - Powder SAP loading: 205 phr — Example 4 - Test duration: 96 hour — Example 4 - Temperature: 200°F — Example 4 - Temperature: 275°F — Example 4 - Brine concentration: 5% brine — Example 4 - Brine concentration: 16% brine — Example 4 - Curing temperature: 330°F — Example 1 - Curing pressure: 70 psi — Example 1 - Curing time: 2.5 hours — Example 1 - Test temperature: 275°F —

Example 1 - Test pressure: about 300 psi — Example 1 - Differential pressure: at least 5000 psi — Example 1 - Hardness: about 86 to 92 Shore A — Example 1 - Powder SAP present: at least about 150 phr — Example 2A - Powder SAP present: at least about 175 phr — Example 2A - Powder SAP present: at least about 197 phr — Example 2A - Powder SAP present: at least about 200 phr — Example 2A - Powder SAP present: at least about 205 phr — Example 2A - Powder SAP present: at least about 225 phr — Example 2A - Heat temperature: about 200°F — Example 3A - Heat temperature: about 275°F — Example 4A - Brine concentration: about 5% weight/volume — Example 5A - Brine concentration: about 16% weight/volume — Example 6A - Swelling: at least about 50% to 90% by volume — Example 7A - Duration: about 2 to 5 days — Example 8A - Pressure resistance: at least about 3000 to 8000 psi — Example 9A - Hydrogenated nitrile elastomer hydrogenation: about 70% to about 91% — synthesis passage - Mooney viscosity ML 1+4 (100°C): about 47 to about 72 — synthesis passage

Relevant Experimental Evidence - Example 1: a swellable packer cured at 330°F and 70 psi for 2.5 h with 16% brine exposure withstood >5000 psi differential pressure. - Example 2: powder SAP loading from 110 to 170 phr was evaluated across 5% and 16% brine at 200°F and 275°F. - Example 3: powder SAP was predicted to reach 75% swell at about 205 phr, while granular SAPs were limited by leaching. - Example 4: 205 phr powder SAP with 10% NBR or 10% HNBR showed strong swell performance and acceptable handling. - Example 10A-12A: claims define nitrile elastomer/silica/SAP formulations capable of maintaining seals at 200°F to 275°F and 5000 psi.

Technical Relevance This patent is relevant to low-acrylonitrile NBR because it shows how nitrile elastomer or HNBR can be formulated into high-performance swellable rubber systems, where low-viscosity nitrile rubber improves processing and downhole performance. It does not teach NBR synthesis, but it strongly informs the application window for low-ACN NBR-derived elastomers.

Patent 4

- Patent Number: US10308807B2
- Patent Title: Dissolvable rubber
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2019
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses degradable elastomeric composites made by combining elastomer particles, including nitrile rubber, with a dissolvable or water-reactive binder such as PVA, PEG, PLA, or related polymers. The examples emphasize molding, dissolution in water or brine, and the formation of seals or other oilfield articles rather than polymerization of NBR itself. - Elastomer content: about 70 vol.% polyurethane rubber particles — Example 5 - Particle size: 30-100 μm — Example 5 - Temperature: about

185°F (85°C) — Example 5 - Dissolution time: about 5 hours — Example 5 - Elastomer content: about 60 vol.% recycled vulcanized rubber particles — Example 6 - Particle size: about 100-1000 μm — Example 6 - Matrix content: about 30 vol.% PEG plasticized PLA — Example 6 - Reinforcement content: about 10 vol.% glass fiber — Example 6 - Temperature threshold: under about 122°F (50°C) — Example 6 - Dissolution window: between about 122°F (50°C) and 212°F (100°C) — Example 6 - Dissolution time: about 30 hours — Example 6 - Elastomer content: about 75 vol.% polyurethane particles — Example 7 - Particle size: about 30-100 μm — Example 7 - Temperature: about 194°F (90°C) — Example 7 - Water exposure time: 24 hours — Example 7 - Trigger time: about 6 hours — Example 7 - Dissolution time: one hour — Example 7 - Elastomer content: about 70 vol.% nitrile butadiene rubber — Example 8 - PVA content: 18 vol.% — Example 8 - Glycerol content: 3 vol.% — Example 8 - $\text{MnCl}_2 \cdot 4\text{H}_2\text{O}$ content: 9 vol.% — Example 8 - Hardness: 78-95 Shore A — Example 8 - Compression set: 5%-60% — Example 8 - KCl concentration: 3% — Example 8 - Temperature range: 130°F to 150°F (54°C to 66°C) — Example 8 - Degradation time: 10-20 hours — Example 8 - Thermoplastic melt range: 300°F to 392°F (150°C to 200°C) — Example 8 - Seal OD: about 4 inches — Example 8 - Seal ID: about 3 inches — Example 8 - Tear strength: 32.1 ± 2.15 kN/m — Example 8 - Tensile strength @ yield: 722.29 ± 23.2 psi — Example 8 - Tensile strength @ break: 807.86 ± 49.31 psi — Example 8 - Tensile elongation @ break: $28.1 \pm 7.71\%$ — Example 8 - Compression set @ 22 h: 75% — Example 8 - Compression set @ 70 h: 78% — Example 8 - Elastomer content: about 60 vol.% nitrile butadiene rubber — Example 9 - PVA content: 16 vol.% — Example 9 - Glycerol content: 8 vol.% — Example 9 - PVP content: 16 vol.% — Example 9 - Hardness: 78-95 Shore A — Example 9 - Compression set: 5%-60% — Example 9 - KCl concentration: 3% — Example 9 - Temperature range: 130°F to 150°F — Example 9 - Degradation time: 10-20 hours — Example 9 - Elastomer content: about 50 vol.% nitrile butadiene rubber — Example 10 - PVA content: 47.5 vol.% — Example 10 - Glycerol content: 2.5 vol.% — Example 10 - Elastomer content: 70 vol.% nitrile butadiene rubber — Example 11 - Glycerol content: 10 vol.% — Example 11 - PVP/ MnCl_2 content: 8 vol.% — Example 11 - Calcium oxide content: 4 vol.% — Example 11 - Hardness: 60-80 Shore A — Example 11 - KCl concentration: 3 vol.% — Example 11 - Temperature range: 130°F to 150°F — Example 11 - Degradation time: 10-20 hours — Example 11

Relevant Experimental Evidence - Example 8: NBR-based dissolvable elastomer degraded 70% in 10-20 h in 3% KCl at 130-150°F. - Example 9: a different NBR/PVA/glycerol/PVP formulation also degraded rapidly in brine. - Example 10: a higher-PVA NBR composite was prepared as a dissolvable elastomeric material. - Example 11: NBR with glycerol, PVP/ MnCl_2 , and calcium oxide degraded into sub-0.5 mm particles. - Example 8: the NBR composite was also demonstrated as a 3D-printable seal material.

Technical Relevance This patent is relevant because it explicitly includes nitrile butadiene rubber in a degradable composite architecture, showing an application path for NBR in dissolvable seals and oilfield tools. It does not address low-acrylonitrile synthesis, but it confirms that NBR can be formulated into functional degradable articles with controlled dissolution behavior.

3. CROSS-PATENT COMPARISON & SYNTHESIS TRENDS

- EP2473279B1 is the only patent in the set that directly modifies nitrile rubber chemistry to lower molecular weight via metathesis and hydrogenation; the others are primarily metathesis-based block-copolymer formation or downstream compounding/application patents.
- EP2473279B1 uses a solution process in monochlorobenzene with 1-hexene, a ruthenium metathesis catalyst, and hydrogenation at 85 bar H₂ and 138°C; EP2057205B1 uses polymer-polymer metathesis in toluene or tetrachloroethane at 95-105°C with Grubbs II or tungsten hexachloride/organotin catalysts.
- EP2473279B1 reports low-Mw HNBR values such as Mn 19,000 g/mol and Mw 34,000 g/mol with PDI 1.78, whereas EP2057205B1 focuses on qualitative evidence of meta-block copolymer formation rather than target molecular-weight reduction.
- US10385647B2 and US10308807B2 both use nitrile elastomer/NBR in formulated articles, but US10385647B2 emphasizes swellable packers with powder SAP at 110-170 phr and 205 phr targets, while US10308807B2 emphasizes dissolvable elastomer composites with NBR at 50-70 vol.% and brine-triggered degradation.
- US10385647B2 includes HNBR and nitrile elastomer as functional components for high-pressure, high-temperature swellable packers; US10308807B2 includes NBR as a degradable particle phase in a water-soluble binder matrix for dissolvable seals.
- Across the set, solution-phase metathesis and catalyst selection are the main chemistry levers, while compounding patents focus on filler/SAP loading, brine exposure, temperature, and pressure resistance rather than monomer feed control or acrylonitrile content control.

4. CONCLUSION

The evidence indicates that the most technically relevant route to low-acrylonitrile NBR performance is not direct acrylonitrile-content adjustment in the extracted patents, but rather post-polymerization molecular-weight reduction and formulation engineering of nitrile rubber/HNBR. EP2473279B1 provides the strongest synthesis guidance: solution metathesis of nitrile rubber followed by hydrogenation can deliver low-Mw, low-PDI HNBR with improved processability, and the disclosed example set supports catalyst loading control and hydrogenation under 85 bar H₂ at 138°C. EP2057205B1 shows that HNBR can also serve as a metathesis substrate for segment interchange, broadening the chemistry toolbox. For application-oriented low-viscosity nitrile rubber, the best-supported formulation window from the evidence is nitrile elastomer/HNBR combined with silica and powder SAP for swellable packers, or NBR combined with water-soluble binders for dissolvable elastomeric articles. Recommended evidence-based synthesis/process parameters are: solution metathesis in chlorobenzene-type solvent, ruthenium-based metathesis catalyst, controlled co-olefin addition where needed, followed by hydrogenation at elevated temperature and hydrogen pressure to obtain low-Mw nitrile rubber feedstocks suitable for downstream compounding.

5. REFERENCES

- EP2473279B1 | Metathesis of a nitrile rubber in the presence of a transition metal catalyst | Not disclosed | EP | 2014 | <https://patents.google.com/patent/EP2473279B1/en>
- EP2057205B1 | Production of meta-block copolymers by polymer segment interchange | Not disclosed | EP | 2017 | <https://patents.google.com/patent/EP2057205B1/en>
- US10385647B2 | Swellable rubber compositions | Not disclosed | US | 2019 | <https://patents.google.com/patent/US10385647B2/en>
- US10308807B2 | Dissolvable rubber | Not disclosed | US | 2019 | <https://patents.google.com/patent/US10308807B2/en>